EPFL - Computer Science

COM-407 Advanced Networks

Lecture 02 — September 29, 2025

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Network Layer IPv4 and IPv6

Lecture Contents - Internet Protocol (IP)

1. The 2 main rules of IP Unicast
2. IPv4 addresses
3. IPv6 addresses
4. Nats
5. Host configuration
6. Hop Limit and TTL
7. ARP (connection with MAC layer)

IP Rule 1: Forward packets according to dest IP address

Why?

Goal of IP = interconnect all systems in the world using IP addresses

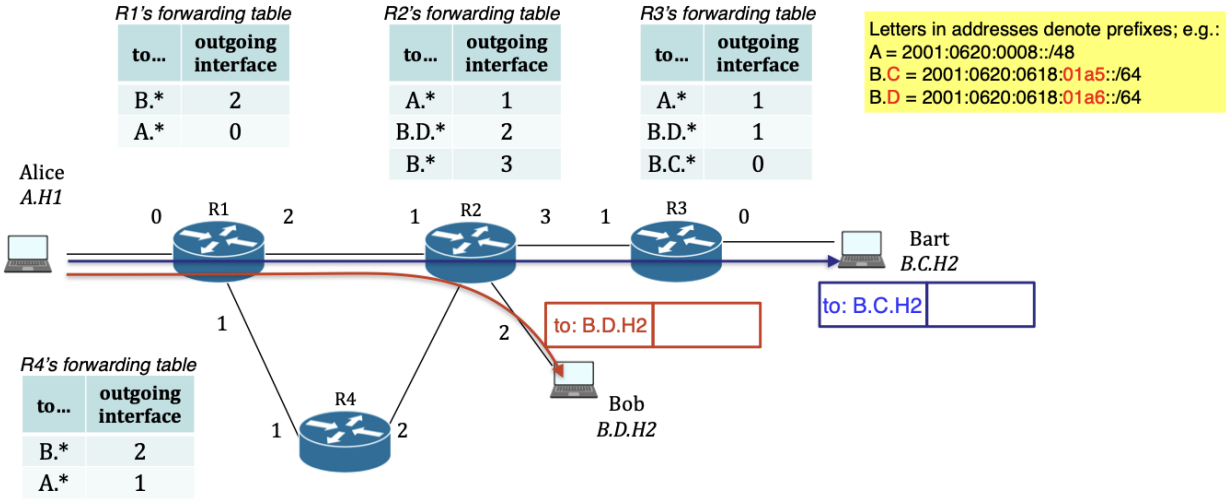
⇒ Each IP address should uniquely identify a system

What?

1. Assign structured addresses:
   - every network interface has an IP address with prefix + suffix: e.g. 128.178.71.202
   - interfaces within a subnet have the same prefix ⇒ same subnet mask
   - but different suffix (a.k.a. host part)
2. Forward packets according to **longest prefix match** principle:
   - every packet contains the destination IP address in its header
   - every system (i.e. **host** = end-system or **router** = intermediate system)
     - has a **forwarding table** (=routing table) and

– forwards each packet based on the **closest** table entry to the destination IP address

Longest prefix match (= closest matching table entry)



- Benefit: addresses can be *aggregated*, tables can be *compressed*

IP Rule 2: Only routers interconnect different LANs/subnets

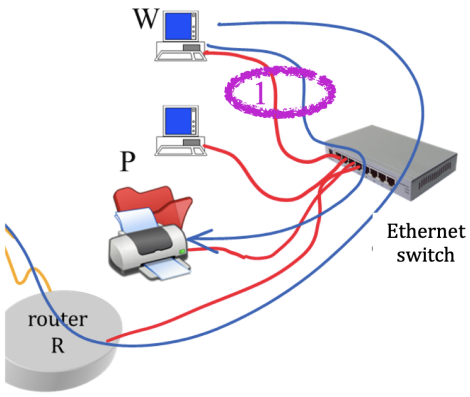
⇒ Between LANs/subnets, use routers

⇒ within each subnet, do not. Instead, use data-link-layer forwarding devices within subnet (e.g. switches, wifi base stations, etc.)

Question

Q: We observe a packet from W to P at 1. Which IP destination address do we see?

1. The IP address of P
2. The IP address of an Ethernet interface of the Ethernet switch
3. There is no destination IP address in the packet since communication is inside the subnet and does not go through a router



S: 1
The IP address is always present in the IP Header, even if communication is inside the same LAN

2. IPv4

Address format: 32 bits, usually written in dotted decimal notation

binary: 32 bits

- example 1: **b**1000 0000 1011 1111 1001 0111 0000 0001
- example 2: **b**1000 0001 1100 0000 1100 1000 0000 0010

dotted decimal: 4 integers (one integer = 8bits/binary digits)

- example 1: 128.191.151.1
- example 2: 129.192.200.2

hexadecimal: 8 hex digits (one hex digit = 4bits/binary digits)

- example 1: **x**80 bf 97 01
- example 2: **x**81 c0 c8 02

Question

Q: The mask 255.255.254.0 means that the subnet is made of the first...

1. 16 bits
2. 18 bits
3. 22 bits
4. 23 bits
5. 24 bits

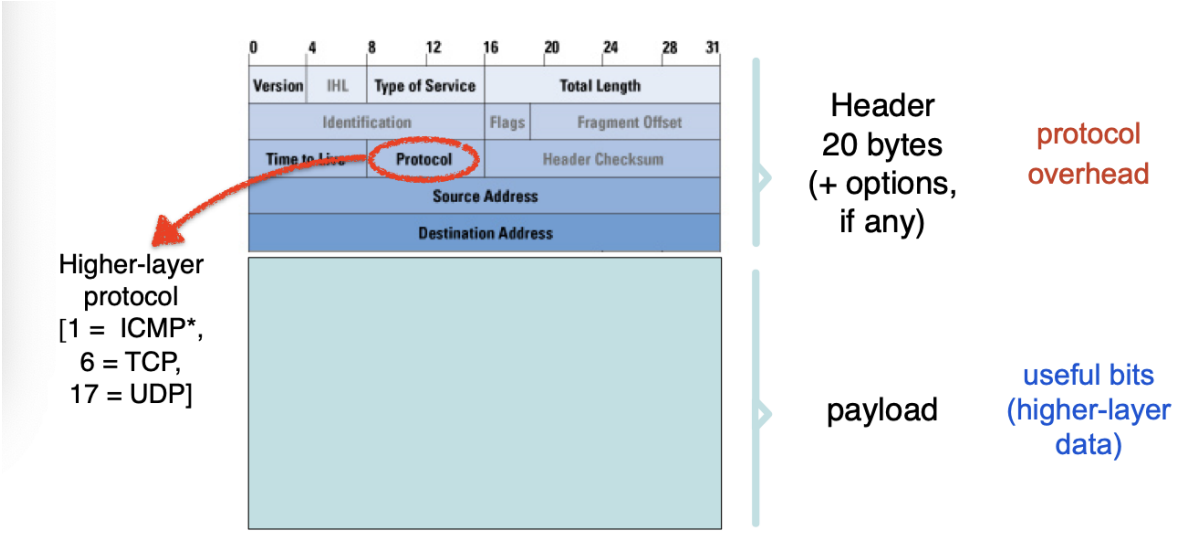
S: 4. 254 is 1111 1110 in binary form $\Rightarrow$ 255.255.254.0 = 1111 1111 1111 1111 1111 1110 0000 0000 in binary form. So, the mask has 23 contiguous bits equal to 1 starting from the beginning.

Reserved IPv4 address blocks

0.0.0.0	**absence** of address, or **any** address
127/8	**loopback** addresses (this host, e.g. 127.0.0.1)
10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	**private** addresses (e.g. at home): used by *anyone*, but *not* in the public Internet (internet routers drop packets destined to such addresses)
100.64/10	private addresses used only by Internet Service Providers (ISPs)— **Carrier Grade NAT addresses**
192.88.99/24	**IPv6-to-IPv4** relay routers
169.254.0.0/16	**link local** addresses (used by systems in the same LAN, assigned automatically with *stateless autoconfiguration*, never routed)
224/4	**multicast**
240/4	reserved "for experimental/future use" until recently
255.255.255.255/32	link local (LAN) **broadcast**

✳ Every other address is *global/public* = *uniquely* identifies a network interface in the public internet

IPv4 Packet Format



✳ (ICMP is used to carry error messages at the network layer)

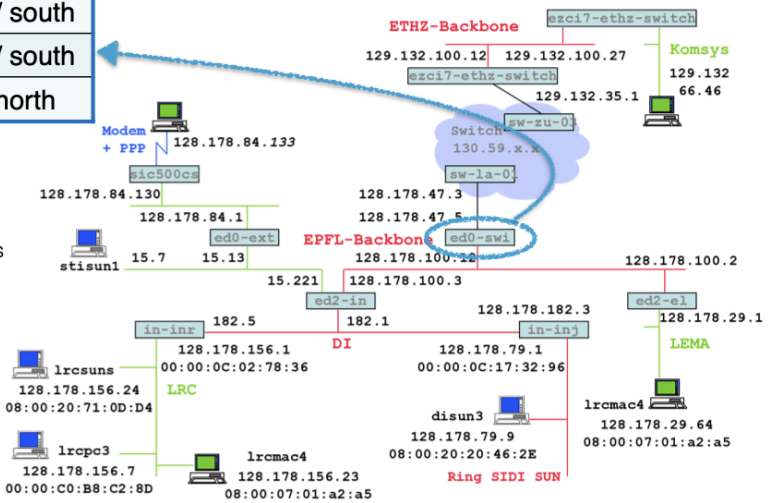
Forwarding table and longest prefix match: example from EPFL

Destination	Next-Hop / Interface
128.178.29/24	128.178.100.2 / south
128.178/16	128.178.100.3 / south
0/0	128.178.47.3 / north

- destination entries in *aggregated/subnet* form
- next hops identified by both "IP" and "interface"
- 0/0 (empty string) = default route for any address

- Longest prefix match means:

```
if packet -> 128.178.*
    if packet -> 128.178.29*
        forward to ed2-el
    else
        forward to ed2-in
else
    forward to sw-la-01
```



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3. IPv6

Why a new version?

IPv4 address space is too small (32 bits $\Rightarrow \approx 4 \cdot 10^9$ unique addresses)

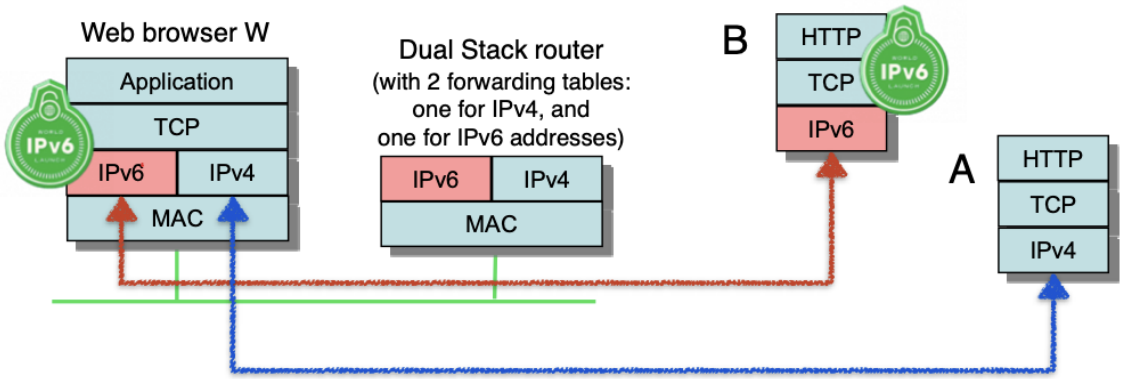
What does IPv6 do?

Redefines packet format with larger addresses of: 128 bits ($\approx 3 \cdot 10^{38}$ unique addresses)

Otherwise, it offers essentially the same services as IPv4

But IPv6 is incompatible with IPv4; routers and host must handle them separately

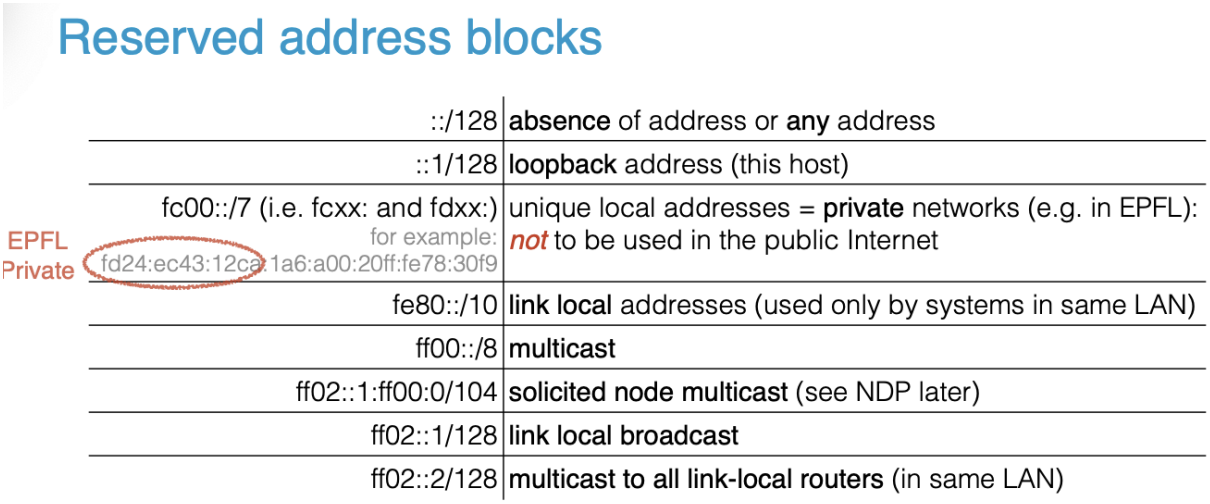
A can talk to W, B can talk to W, A and B cannot communicate at the network layer



IPv6 Address format

- Hex notation with lowercase letters
  - 8 hextets separated by ":"
  - **hextet** = [0-4] hex digits = 16 bits
- Compression Rules
  - leading 0s of a hextet can be omitted
  - :: replaces any number of 0s in more than one hextet; used at most once without ambiguity
  - 2002:0000:0000:0000:0000:ffff:80b2:0c26 ⇒ 2002::ffff:80b2:c26
  - 2001:0620:0618:01a6:0000:20ff:fe78:30f9 ⇒ 2001:620:618:1a6:0:20ff:fe78:30f9

Reserved address blocks



Note: IPv6 private prefixes are also allocated, not reused (in the public internet))

an EPFL public address: 2001:620:618:1a6:a00:20ff:fe78:30f9

an EPFL private address: fd24:ec43:12ca:1a6:a00:20ff:fe78:30f9

fd denotes a private address. The red sequence is something specific to EPFL. This is not meant to be reused by someone else. Instead of IPv4 where 192.168.X.X are private addresses for everyone...

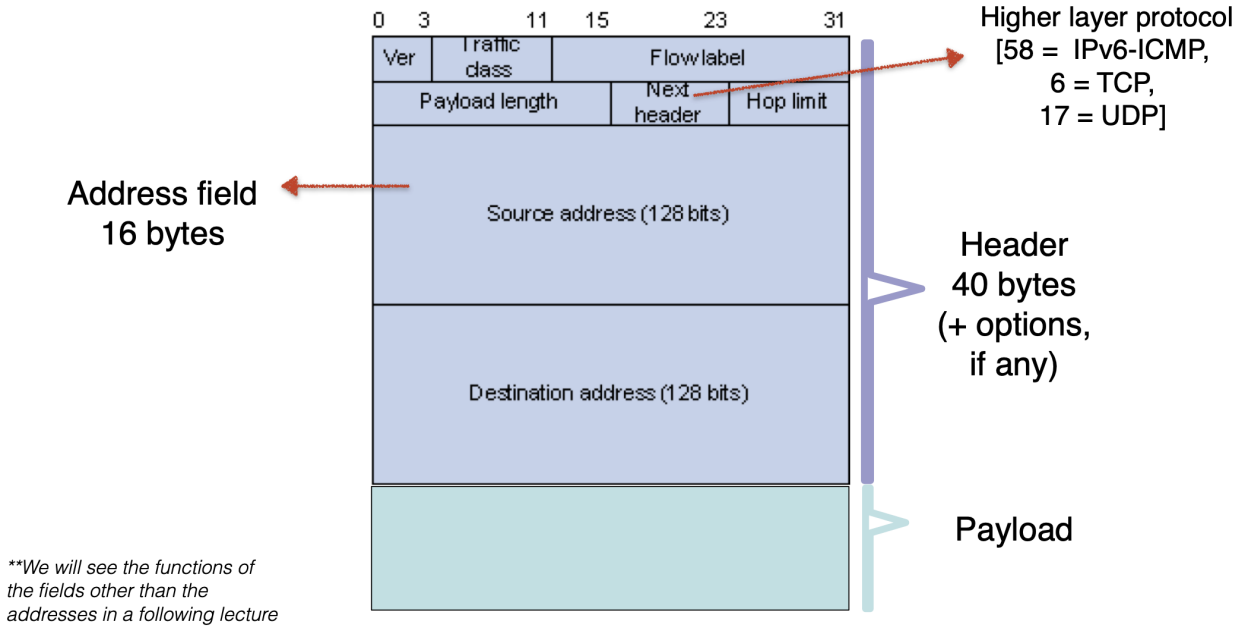
A few IPv6 global unicast addresses

- Currently, only the block 2000/3 (i.e. 2xxx and 3xxx) is for global/public unicast addresses.

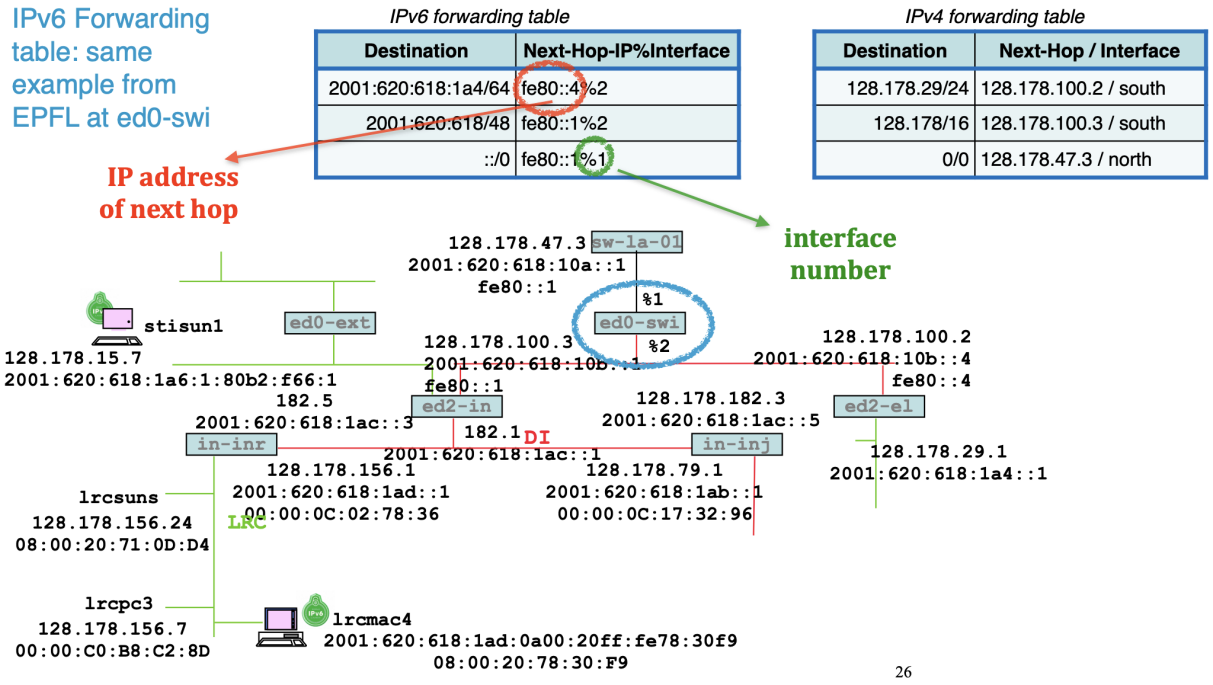
2001:620::/32	Switch
2001:620:618::/48	EPFL
2001:620:8::/48	ETHZ
2a02:1200::/27	Swisscom
2001:678::/29	provider independent address
2001::/32	Teredo (tunnels IPv6 in IPv4)
2002::/16	6to4 (tunnels IPv6 in IPv4)

- Hierarchical allocation: Networks served by a provider use blocks that are subnets of the provider's address block (e.g. check EPFL, ETHZ and SWITCH)

IPv6 Packet Format



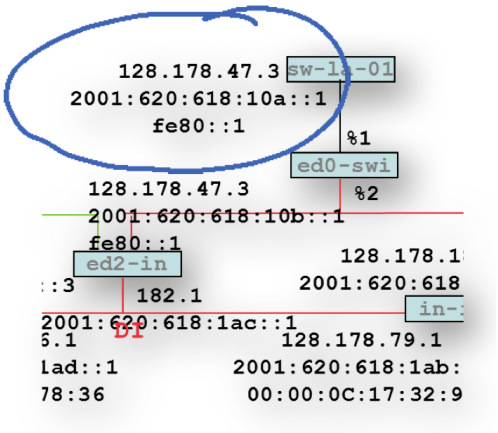
Note: IPv6 is not always a good idea because of big header size (40 bytes) which can make for lower throughputs. For instance, for a IoT setup (multiple physical devices sending data frequently), IPv6 may not be suitable.



Multiple addresses per interface are the rule with IPv6

A host interface typically has

- one or several link local addresses
- plus one or several global unicast addresses (secure (GCA) address, temporary addresses)



The preference selection algorithm, configured by OS, says which address should be used as source address - see RFC 3484

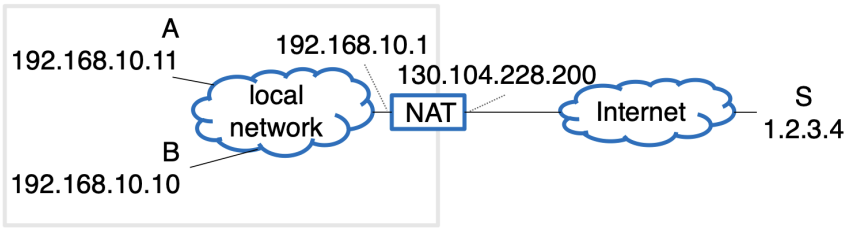
In contrast, in IPv4: there is usually only one IP address per interface

The zone index identifies an interface inside one machine that has several interfaces (e.g. a router)

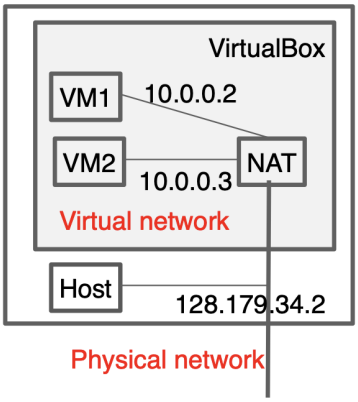
- typically visible in Windows machines
- never inside an IP packet

E.g. fe80::1%2 means: the destination IPv6 address fe80::1 on interface %2

4. NAT (Network Address Translation) box



- Why invented? To allow $n > 1$ devices to share a single public IP address; e.g.:
  - Internet service provider gives you a single IPv4 address, but you have n devices at home and need more addresses.
  - A virtualization platform offers n guest VMs in one host and allows them to communicate with outside, using the single IP address of the physical machine



- What does it do?
  - NAT translates (= masquerades) an internal IP address and internal port number into NAT IP address and NAT port
  - Internal addresses are typically **private**, ports are either UDP or TCP
  - From outside, one sees only the (**public**) NAT IP address and a NAT port

- NAT is a network-layer middle box, but violates
  - layering ⇒ manipulates 2 layers to work (IP and port)
  - IP rule that public addresses should identify host uniquely

Note: Also used in IPv6, but very often used for IPv4. Designed for a temporal solution because of missing public addresses from IPv4